Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/01/2007
PI Name:	Larson, Melora E. Ph.D.		
Project Title:	Experiments Along Coexistence Near Tricriticality (EXACT (2000))		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	818-354-8751
Organization Name:	NASA Jet Propulsion Laboratory		
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PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 06/30/2007 per D. Strayer (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nash, Alfred (Jet Propulsion Laboratory) Mulders, Norbert (University of Delware) Vollmayr-Lee, Ben (Bucknell University)		
Grant/Contract No.:	GSN 4534		
Performance Goal No.:	1.24		
Performance Goal Text:	1.24 To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	EXACT 2000 will measure the second sound velocity as a function of concentration and temperature in mixtures of helium-3 and helium-4 near the tricritical point. From these measurements, the superfluid density will be derived, and the critical behavior of the superfluid density will be found. EXACT 2000 will also use these measurements to measure the shape of the coexistence curve and the Lambda-line as a function of temperature and concentration. The task objective of EXACT 2000 is to make high resolution measurements on mixtures of 3He and 4He to determine the critical exponents at the tricritical point. This task will improve upon existing measurements at the tricritical point by an order of magnitude. These measurements will then be compared to existing, exact theoretical predictions for any possible discrepancy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research tests one of the most fundamental theories about how matter behaves when it changes state. Since industry often makes use of how matter behaves as it changes state, any better understanding of this area of physics has potential impact back on industry.
Task Progress:	The past few years of the EXACT project were spent on improving the experimental apparatus with the goal of taking additional measurements of the phase separation of mixtures of helium-3 and 4. Unfortunately, we were unable to complete all the modifications to the apparatus before the funding was exhausted. We did have two additional summer students (one new, and one returning) work over the summer of 2006. Both of these students were part of the Caltech summer student fellowship program. As part of that program, they both wrote final reports which were released within the Caltech community.
Bibliography Type:	Description: (Last Updated: 10/02/2007)